

# GOKUL GOPALAKRISHNAN

gokulg846@gmail.com | ggopala@purdue.edu | +1 (765) 543-8560 | West Lafayette, IN | [linkedin.com/in/gokulgopal/](https://www.linkedin.com/in/gokulgopal/)  
**SUMMARY**

- **Hardware Program Management:** Experience managing mechanical engineering testing and development programs.
- **Six Sigma Yellow Belt certified**, experience with **product improvement, statistical analysis, and problem-solving**.
- Mechanical Engineer with strong hands-on experience and technical skills looking for full-time roles starting May 2026.

## EDUCATION

**Purdue University** West Lafayette, IN  
*Master of Engineering Management (Focus: Industrial Engineering)* Aug 2023 – May 2026

**Courses:** Simulation Design and Analysis with MATLAB, Artificial Intelligence, Programming with Arduino

**SASTRA University** Thanjavur, India  
*B. Tech in Mechanical Engineering* Jul 2017 – Jun 2021

**Courses:** Microcontrollers & Embedded systems, Control systems, Robotics, Design for Manufacturing

## WORK EXPERIENCE

**Product Engineer Co-op** Columbus, IN  
*Cummins* Feb 2025 – Jan 2026

- Spearheaded the investigation into bolted joint failures from **field returns**, developing a comprehensive shaker test plan using field-derived PSD profiles to validate design fixes, mitigating potential warranty exposure on 5000+ units.
- Supported engine emissions validation for new products, defining end-to-end test strategies, coordinating with stakeholders, ensuring test accuracy and repeatability to support production readiness and design decisions.
- **Designed production-scale components in PTC Creo**, applying GD&T, tolerance stack-ups, and DFM/DFA principles to ensure manufacturability in high-volume production.
- Performed **FEA using ANSYS Discovery** to assess **back pressure tradeoffs** across multiple design concepts, iterating to improve performance of existing model, achieving 20% performance improvement on two potential concepts.
- Led **inventory management, working cross-functionally** with testing, quality, procurement, & fabrication teams.
- Migrated legacy test-vehicle **data analysis workflows** from Excel to MATLAB, **cutting manual effort by 20+ hrs/week**.
- Deployed **MATLAB-based** data analysis and validation scripts for manufacturing & testing data, generating automated data visualizations, improving data interpretability, enabling informed engineering decisions.

**Student Consultant** Remote  
*Purdue-Bayer Innovation* Aug 2023 – Dec 2023

- Led process optimization for a Nigerian Agri-Tech startup, reducing **operational costs by 45%**.
- Built a **Computer Vision tool with PyTorch** to automate farmland mapping from satellite imagery.

**Analyst** Bengaluru, India  
*Accenture* Jul 2021 – Jul 2023

- **Built real-time monitoring and predictive maintenance models** for industrial IoT systems using Python and SQL.
- Integrated high-frequency sensor data pipelines to detect anomalies, enabling early indicators of component failure.
- Designed **3D CAD models in SolidWorks** to build accurate Digital Twin representations of physical assets, enabling high-fidelity virtual simulation of system behavior.
- Created Tableau dashboards and Gantt charts to track KPIs, boosting **SLA compliance by 12%**.

## PROJECTS

**Student Lead - Powertrain, SASTRA Formula Bharat Team**

- Led **powertrain integration**, coordinating mechanical, electrical, and control elements to meet performance requirements.
- Modeled the powertrain subsystem using **Solidworks**, performed FEA simulations using ANSYS Fluent.
- Worked with PLCs and electronics using C/C++, Arduino and Raspberry Pi.
- Utilized agile, **Six Sigma methodologies**, performed PFMEA to ensure proper design and development.

**Graduate Data Science Researcher – The Data Mine at Purdue**

- Built **end-to-end ML workflows** with PyTorch. Applied **XGBoost** to boost pharma marketing conversion by 15%.
- Conducted **sensitivity analysis** to determine key model drivers influencing conversion outcomes.

## SKILLS

- **Mechanical Engineering:** PTC Creo, Solidworks, NX, Windchill, GD&T, DFM/DFA, FEA (ANSYS Discovery/Fluent).
- **Data & Simulation:** Python, MATLAB/Simulink, SQL, Minitab, Root Cause Analysis.
- **Hardware & Prototyping:** Arduino, Raspberry Pi, IoT, Six Sigma (Yellow Belt), Design of Experiments.